

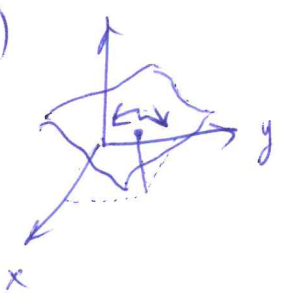
1. Ders

Gradient $\vec{\nabla}$: operator.

Suppose we have a scalar function $\varphi(x, y, z)$, which depends on position vector.

If we interested in steepness of φ in each coordinate system direction.

(2D)



$$\vec{\nabla} \varphi(x, y, z) \equiv \frac{\partial \varphi(x, y, z)}{\partial x} \hat{x} + \frac{\partial \varphi(x, y, z)}{\partial y} \hat{y} + \frac{\partial \varphi(x, y, z)}{\partial z} \hat{z}$$

$\varphi(x, y, z) \rightarrow \boxed{\vec{\nabla}} \rightarrow$ vector function $\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$

scalar function

$$\boxed{\vec{\nabla} \varphi = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \varphi = \hat{x} \frac{\partial \varphi}{\partial x} + \hat{y} \frac{\partial \varphi}{\partial y} + \hat{z} \frac{\partial \varphi}{\partial z}}$$

Ex

$$\varphi = |\vec{r}|^2 = x^2 + y^2 + z^2$$

$$\vec{\nabla} \varphi = 2x \hat{x} + 2y \hat{y} + 2z \hat{z} = 2(x \hat{x} + y \hat{y} + z \hat{z}) = 2\vec{r}$$

$$\vec{\nabla} \varphi = \vec{\nabla} r^2 = 2\vec{r}$$

Ex $\vec{\nabla} r = ?$ $\vec{\nabla} (x^2 + y^2 + z^2)^{\frac{1}{2}} = \left(\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{1}{2}} \cdot 2x \right) \hat{x} + (\dots) \hat{y} + (\dots) \hat{z}$

$$= \frac{x \hat{x}}{r} + \frac{y \hat{y}}{r} + \frac{z \hat{z}}{r} = \frac{1}{r} (x \hat{x} + y \hat{y} + z \hat{z}) = \frac{\vec{r}}{r} = \frac{r \hat{r}}{r}$$

$$\boxed{= \hat{r}} \quad (9)$$

ex) $\vec{A} = r (\cos \theta \hat{y} - \sin \theta \hat{x})$

$$\vec{A} = r \cos \theta \hat{y} - r \sin \theta \hat{x}$$

$$A = x\hat{y} - y\hat{x}$$

$$\vec{\nabla} \cdot \vec{A} = 0 + 0 + 0 = 0$$

Curl $\vec{\nabla} \times$

2. Ders

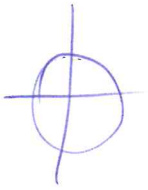
$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Ex) ~~$\vec{\nabla} \times \vec{r}$~~ $\vec{\nabla} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0$

vector $\vec{\nabla} \times$ vector f.

Bx) $\vec{A} = -y\hat{x} + x\hat{y}$ $\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 0 \end{vmatrix} = \hat{x} \cdot 0 + \hat{y} \cdot 0 + \hat{z} (1-1)$

$= 2\hat{z}$



Show that.

Exercise

1) $\vec{\nabla} (\vec{A} \cdot \vec{B}) = \vec{\nabla} (\vec{B} \cdot \vec{A}) = (\vec{B} \cdot \vec{\nabla}) \vec{A} + (\vec{A} \cdot \vec{\nabla}) \vec{B} + \vec{B} \times (\vec{\nabla} \times \vec{A}) + \vec{A} \times (\vec{\nabla} \times \vec{B})$

2) $\vec{A} \times (\vec{\nabla} \times \vec{B}) = \nabla (\vec{A} \cdot \vec{B}) - (\vec{A} \cdot \vec{\nabla}) \vec{B}$

$$\nabla V = \frac{\partial V}{\partial x} \hat{x} + \frac{\partial V}{\partial y} \hat{y} + \frac{\partial V}{\partial z} \hat{z}$$

$$= (-F_x) \hat{x} + (-F_y) \hat{y} + (-F_z) \hat{z}$$

$$\boxed{\nabla V = -\vec{F}}$$

Divergence Operator : $\vec{\nabla}$ dot operator.

Consider a vector that depends on position (x, y, z) , say $\vec{A} = \vec{A}(x, y, z)$
 $= \vec{A}(\vec{r})$

$\frac{\partial A_i}{\partial x_i}$ for $i=1,2,3$ are summed yields $\vec{\nabla} \cdot \vec{A}$: "divergence of $\vec{A}$ "

dot sign.

$$\text{Then } \vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (A_x, A_y, A_z)$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Vector function $\rightarrow \boxed{\vec{\nabla} \cdot}$ $\rightarrow$ scalar function.

ex)

$$\vec{\nabla} \cdot \vec{r} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (x, y, z) = 1 + 1 + 1 = 3 \neq 0$$

ex)

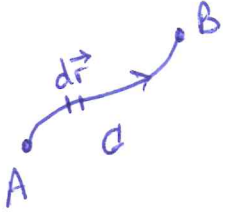
$$\vec{A} = y \hat{y}$$

$$\vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (0, y, 0) = 1 \neq 0$$

Rule of Change of derivative related vectors. $\vec{\nabla} \phi$ $\vec{\nabla} \cdot \vec{A}$ $\vec{\nabla} \times \vec{A}$

Vector Integration:

I) Line Integral.



C : Well defined contour starting at point A, ending B.

Along this path, we are given a quantity that is a function of points (on the contour).

$\int_C d(\text{quantity of interest})$ is called a line integral.

3 main types of line integral.

i) $\int_C \phi d\vec{r}$

ii) $\int_C \vec{V} \cdot d\vec{r}$

iii) $\int_C \vec{V} \times d\vec{r}$

$$d\vec{r} = \hat{x} dx + \hat{y} dy + \hat{z} dz$$

$$I = \int_C \phi d\vec{r} = \hat{x} \int_C \phi(x,y,z) dx + \hat{y}(\dots) + \hat{z}(\dots)$$

Note:

We assumed $\int_C \hat{x} \phi dx = \hat{x} \int \phi dx$ which is correct in cartesian coord system.